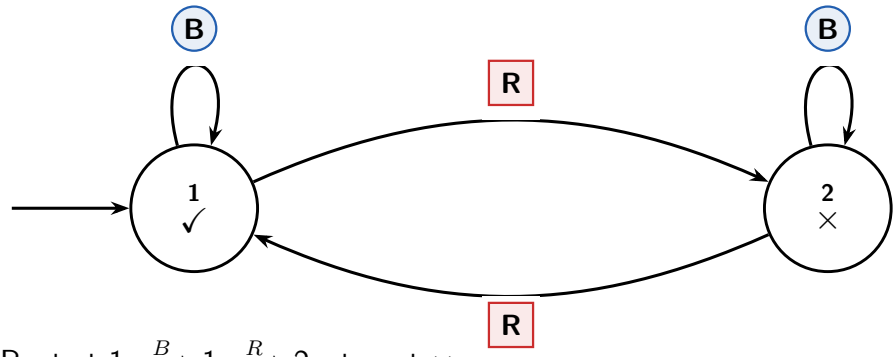
Start at the small arrow with one marker. Read left to right. Follow one matching arrow for each card, then hide the card. Each circle has one arrow for each kind of card and a fixed ✓ or ×.

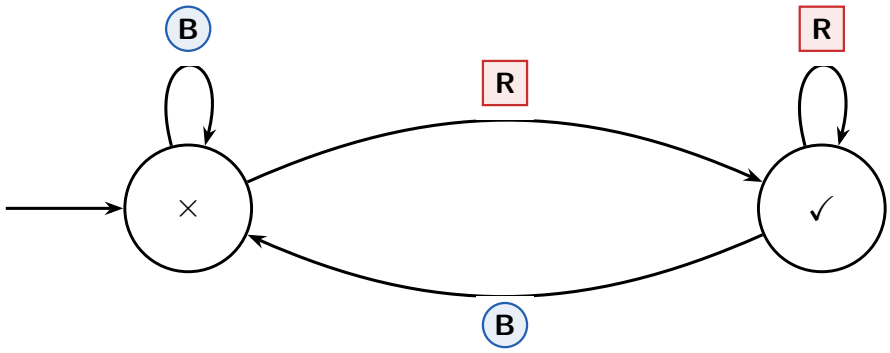
Problem 1: Try each row, starting fresh each time, and circle its answer. Make three different six-card rows that finish at ✓.



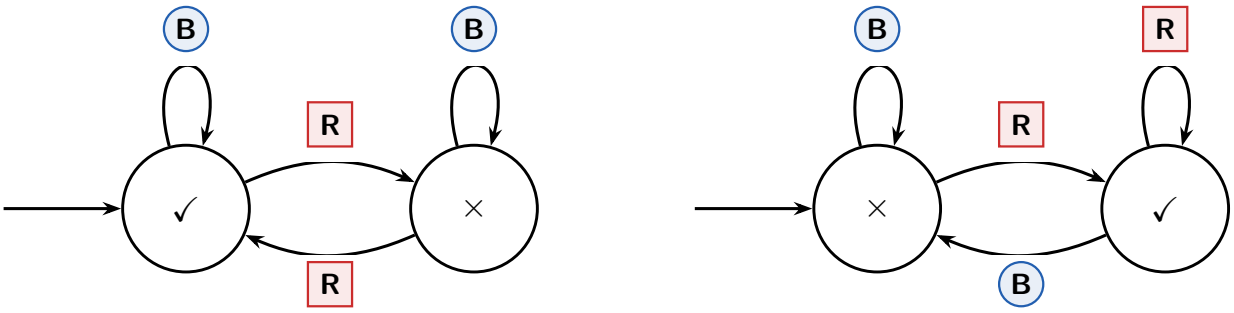
Example B R: start 1 $\xrightarrow{B}$ 1 $\xrightarrow{R}$ 2; stop at ×.

no cards	✓	×
B	✓	×
R B R	✓	×
R R R	✓	×
R R R R	✓	×
B B B R B	✓	×

Problem 2: Make every four-card row that finishes at ✓. Show each row once.



Problem 3: Make six different rows that give these machines different answers. Use six cards in each row.

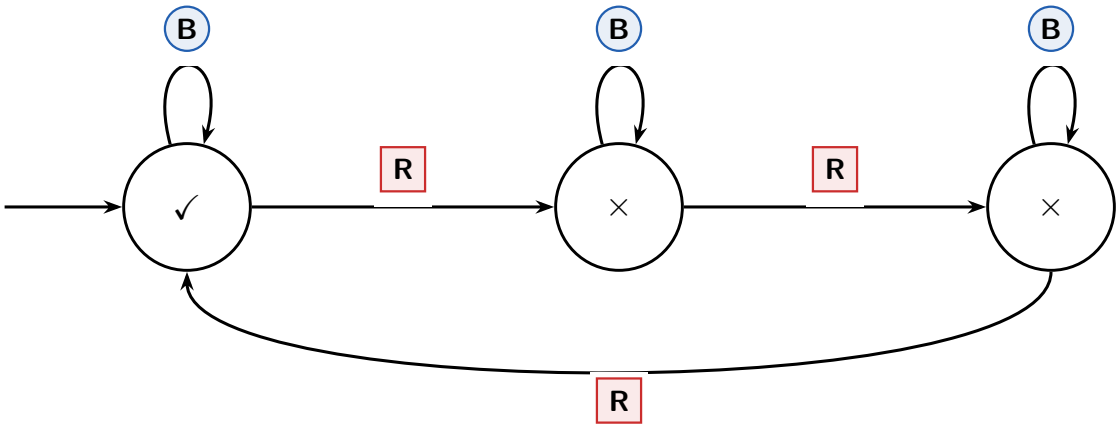


Problem 4: Use two circles to make a machine that says ✓ exactly when the last card is blue. With no cards, it must say ×.



Problem 5: Make a machine that says ✓ when no red cards appear and × when any red card appears. Use as few circles as you can.

Problem 6: For each pair, can you add the same cards in the same order after both printed rows to get ✓ and ×? Adding no cards is allowed.



no cards	+	✓	×
R		✓	×
R	+	✓	×
R R		✓	×
R R	+	✓	×
R B R		✓	×
B	+	✓	×
R R R		✓	×